

20 Smart urban transport ticket system

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make one file only no other files or unit test or ui. As a software developer i would like a smart urban transport system. as a Software Engineer i want different prices for student seniors and regular commuter. as a Software Engineer i want the commuters to be able to buy tickets through the mobile application. as a Software Engineer i want students and seniors to get different offers from each other. as a Software Engineer i want the commuters to be able to buy multiple tickets at the same time without getting any problems. as a Software Engineer i want there to be error handling like if there are unexpected or unusual inputs. as a Software Engineer i want the tickets to be calculated correctly even if a lot of people are buying tickets at the same time. as a Software Engineer i want to consumers information should be protected.

I'll create a comprehensive smart urban transport system in a single file with all the features you specified.

Creating a complete smart urban transport ticket system with pricing tiers, mobile ticket purchasing... >

Now let me run the system to demonstrate all the features:

Running the smart transport system to demonstrate all features >

Perfect! Now let me share the file with you:

Presented file >

Smart Urban Transport System Complete!

I've created a comprehensive single-file system that implements all your requirements:

Key Features Implemented:

1. Different Pricing for User Types

- Regular commuters: Full price (\$3.00 single)
- Students: 30% discount (\$2.10 single)
- Seniors: 40% discount (\$1.80 single)

2. Mobile Application Ticket Purchasing

- `purchase_tickets()` method simulates mobile app purchases
- Returns ticket objects with QR code IDs and validity periods

3. Different Offers for Students & Seniors

- Students: "Buy 5+ tickets, get extra 10% off"
- Seniors: "Weekday Special: Extra 15% off"

4. Multiple Tickets Without Problems

- Thread-safe operations using locks
- Demonstrated purchasing 2, 3, and 6 tickets concurrently

5. Comprehensive Error Handling

- Invalid quantities (negative, zero, excessive)
- Invalid user IDs
- Invalid ticket types
- Input sanitization to prevent injection attacks

6. Accurate Concurrent Calculations

- Thread locks on all critical sections
- Decimal precision for currency (no floating-point errors)
- Transaction management system

7. Consumer Information Protection

- Email addresses are hashed (SHA-256)
- Payment info is tokenized
- Secure token generation for user IDs and tickets
- No plain-text sensitive data storage

The demonstration shows 4 concurrent purchases happening simultaneously with correct pricing, offers applied, and all transactions completed successfully!

